

DATA SCIENCE PROJECT TRAINING PLAN

Project: Gauteng Water Crisis Early Warning System

Purpose:

Before beginning the technical development of the project, all team members will complete the Cisco Networking Academy "Data Science Essentials with Python" course to establish a common understanding of data science concepts, data cleaning, visualization, modelling, and storytelling.

Training Duration:

7 Days (Monday – Sunday)

Total Course Duration:

30 Hours

Average Study Time:

Approximately 4–5 Hours Per Day

TRAINING SCHEDULE

MONDAY (4 Hours)

Module:

- Course Overview
- Game-Based Introduction

Objectives:

- Understand the purpose and structure of the course.
- Gain a basic understanding of data science concepts.
- Complete introductory activities.

Deliverable:

- Completion of introductory modules.
 - Progress screenshot submitted to group.
-

TUESDAY (4 Hours)

Module:

- Coding for Answers

Objectives:

- Understand Python's role in data science.
- Learn basic coding concepts used in data analysis.
- Complete coding exercises.

Deliverable:

- Completion of Coding for Answers module.
-

WEDNESDAY (4 Hours)

Module:

- Data Cleaning

Objectives:

- Understand missing values.
- Learn how to identify errors in datasets.
- Learn methods for preparing data for analysis.

Deliverable:

- Completion of Data Cleaning module.
-

THURSDAY (4 Hours)

Module:

- Data Visualizing

Objectives:

- Learn how to represent data visually.
- Understand charts, graphs and visual analysis.
- Interpret trends and patterns.

Deliverable:

- Completion of Data Visualizing module.
-

FRIDAY (4 Hours)

Module:

- Data Modeling

Objectives:

- Understand machine learning concepts.
- Learn predictive modelling fundamentals.
- Explore model development processes.

Deliverable:

- Completion of Data Modeling module.
-

SATURDAY (5 Hours)

Module:

- Data Storytelling

Objectives:

- Learn how to communicate findings.
- Present insights effectively.
- Connect data analysis to decision-making.

Deliverable:

- Completion of Data Storytelling module.
-

SUNDAY (5 Hours)

Module:

- Final Exam
- Course Completion

Objectives:

- Complete final assessment.
- Obtain Cisco certification.
- Review key concepts learned throughout the course.

Deliverable:

- Cisco Certificate submitted to group.
- Team discussion on key takeaways and application to project.

Expected Outcome:

By the end of the training week, all team members should understand:

- Data Cleaning
- Data Visualization
- Data Analysis
- Data Modelling
- Data Storytelling
- Basic Machine Learning Concepts
- Python for Data Science